

Building a Roster: Positional Value, the Cap, Free Agency, and the Draft

How X's and O's turn into money: which players are worth paying, how a contract really hits the cap, where players come from, and why the same lineman is worth more to one team than another

The broadcast cuts to a graphic during a stoppage: a quarterback's face, and under it "**Cap hit: \$55.4 million.**" A few minutes later the color analyst mentions that the same team "can't afford" to keep its second-best receiver next spring, and that the backup quarterback who just came in "was on the practice squad on Tuesday." Every one of those sentences is about the business underneath the scheme. In May 2025 the San Francisco 49ers gave Brock Purdy a contract worth up to \$265 million. Three seasons earlier he was the last player chosen in the draft, and for those three seasons he cost the team less than one percent of its salary cap a year while taking it to an NFC Championship Game and a Super Bowl. Why was the cheap version of Purdy the most valuable asset the 49ers owned, and why did they happily make him one of the most expensive? This chapter answers that, and the questions under it: which positions are worth paying, how a contract really counts, where players come from, and why a lineman one team passes on is exactly what another team was looking for.

You'll be able to:

- Rank the positions by **value** using contract and play-by-play data, and explain why a quarterback's play predicts winning while a team's spending on any position barely does.
- Read a contract the way a cap analyst does: turn a headline number into cap hits, guarantees and **dead money**, and say what a restructure or void years do to next year's cap.
- Explain how teams acquire players (free agency, the franchise and transition tags, compensatory picks, trades before the deadline, waivers) and what each route costs.
- Price a draft pick two ways, with the Jimmy Johnson chart and with a surplus-value curve, and explain why they disagree and when the disagreement doesn't matter.
- Say what scouts grade at each position, what the combine and **RAS** can and can't tell you, and why the same prospect is a better **scheme fit** for one team than another.
- Follow a team's roster moves during the season: the 53, the 48 who dress, practice-squad elevations, injured reserve and the emergency third quarterback.

You'll need

- **Expected Points, EPA and Success Rate: EPA** (expected points added: how much a play changed the offense's expected score). This chapter uses EPA per dropback as its measure of how well a team passed, and leans on 13-02's finding that passing is worth more than running.
- **The Game Plan: A Week Inside an NFL Building**: the **game plan** and the **matchup** hunting that a roster has to make possible. A game plan can only use the players the front office bought.
- **The 1978 Liberation** (optional): where free agency (1993) and the **salary cap** (1994) came from. The cap is a hard limit, the same for every team, on what a club may count against its payroll in a year.

The problem: 53 players, one budget, and a draft

An NFL general manager has three constraints that no fantasy league or Madden franchise mode quite captures.

The cap is hard. Every team gets the same salary cap, set each year from a share of league revenue: \$279.2 million for 2025 and \$301.2 million for 2026.¹ Unlike baseball or the NBA, there is no luxury tax that lets a rich owner pay his way over it. A dollar spent on one player is a dollar that can't be spent on anyone else, so every contract is a decision about the whole roster.

¹2025 cap: NFL announcement, February 2025 ("NFL salary cap set at record \$279.2 million for 2025 season," NFL.com). The earlier caps used in this chapter's code (2011–2024) are the NFL's announced figures as listed by Over the Cap, <https://overthecap.com/salary-cap-space>. The 2026 figure of \$301.2 million should be checked against the NFL's 2026 announcement.

The roster is big and the game is specialized. A team carries 53 players, dresses 48 on game day, and needs starters at roughly 24 different jobs once you count the kicker, punter and long snapper. A left tackle can't play cornerback. So the budget can't simply follow "buy the best players"; it has to buy a *team*, with a backup at every spot the season will expose.

Most talent arrives through a draft you don't control. The worst teams pick first, and rookies are paid on a fixed scale. That makes the draft the cheapest source of good players, and it means a team's cost of talent depends on *when* it acquired a player as much as on how good he is.

Put those together and the GM's real question isn't "who is good?" It is "**where does the next dollar, or the next pick, buy the most winning?**" The rest of this chapter is the toolkit for answering it.

! Why it exists: the salary cap

The cap is a deal between the owners and the players' union, renewed in each **collective bargaining agreement** (the current one, signed in 2020, runs through the 2030 season). The players get a fixed share of league revenue, about 48 percent, paid out through the cap and benefits; the owners get cost certainty and a league in which a small-market team can afford the same roster as a big-market one. The trade-off is the one this chapter keeps returning to: under a hard cap, overpaying anyone is never free, because someone else on the roster pays for it.

i Madden vs. real life

In Madden's franchise mode, cutting an expensive veteran usually frees his whole salary. In the NFL, a release can *cost* cap space this year, because of the bonus money already paid (the dead money section below shows why). And no real GM can trade three backups for a star the way the game's AI often accepts: picks and players are priced by charts every front office knows, and the other team's GM has read this chapter too.

Positional value: not every position is worth the same

Start with what teams actually pay. Figure 1 shows the ten richest active contracts at each position, measured the way cap analysts compare deals across years: average salary per year (**APY**, the contract's total value divided by its years) as a share of the salary cap in the year the deal was signed.

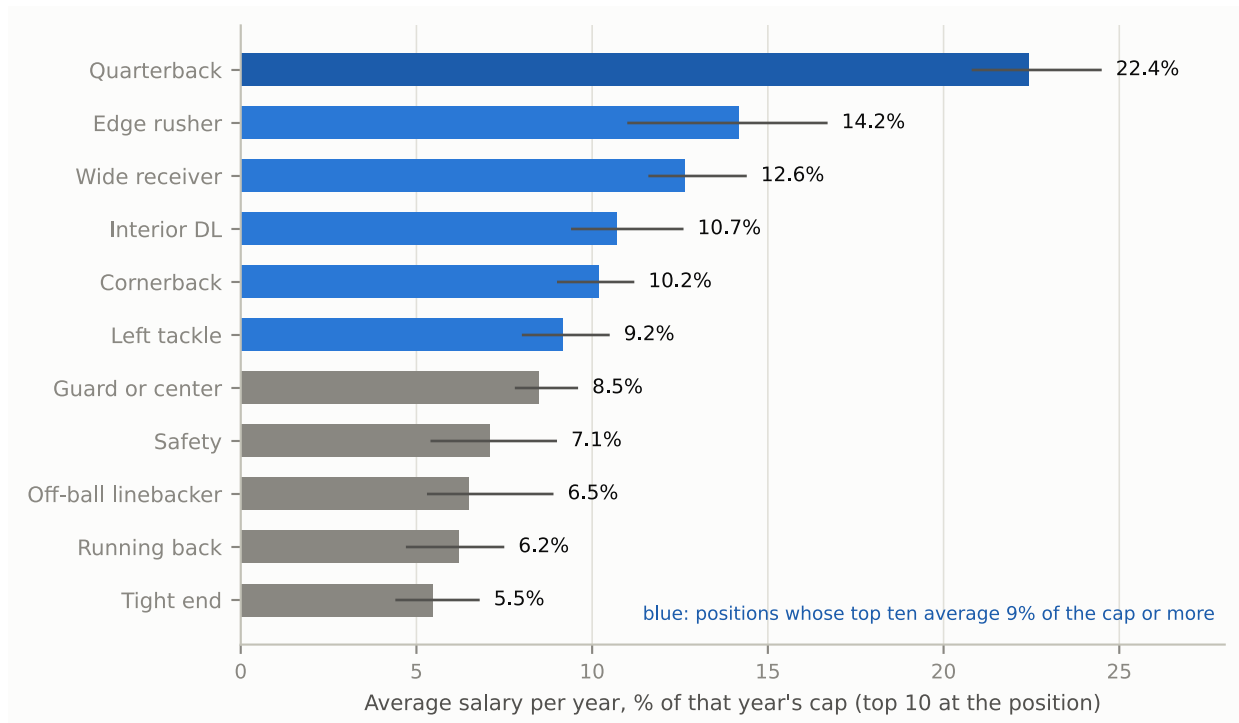


Figure 1: The market's verdict on positional value. Each bar is the average of the ten richest active contracts at that position as of the 2026 season, measured as average salary per year (APY) divided by the salary cap in the year the contract was signed; the thin line runs from the tenth-richest to the richest deal. A top-ten quarterback is paid about one and a half times what a top-ten edge rusher is and more than three times what a top-ten running back is. Data: Over the Cap contract data via nflverse (load_contracts).

That chart is **positional value** as the market sees it: how much a *position*, not a particular player, contributes to winning, and so how much it's worth paying for. Quarterback is alone at the top, at 22.4% of the cap for a top-ten deal. Then come the **premium positions**: edge rusher and wide receiver, then a cluster of interior defensive linemen, cornerbacks and left tackles at 9 to 11 percent. Below them come guards and centers, then safeties, off-ball linebackers, running backs and tight ends, all between 5 and 7 percent. (Interior defensive line is high because the market pays a few rare interior pass rushers like edge rushers; a run-stuffing tackle is paid far less.) Each step has a reason.

Quarterback touches every snap. He decides where the ball goes on every pass and many runs, and Expected Points showed that passing plays carry more value per snap than runs and swing far more widely. A position that controls the highest-leverage action on every play is worth more than one that matters on a third of them.

The premium group decides the passing game. An edge rusher who wins one-on-one ends drives with sacks and hurried throws; the left tackle (for a right-handed quarterback) is the man assigned to stop him from the quarterback's blind side, the side he can't see; a receiver who wins against man coverage gives the quarterback somewhere to throw; and a cornerback who can cover him alone frees the rest of the defense to help elsewhere. Every one of those players wins or loses a matchup on most passing downs, and they are scarce: you can't develop a 6-foot-5, 265-pound man who runs a 4.5 forty from the seventh round.

The bottom group is either replaceable or rarely decisive. Running back is the classic case. Most of a run's yards are created by the blockers and the scheme before the back touches the ball (13-02's rush-yards-over-expected showed how little separates most backs once you account for the blocking), backs wear down quickly, and the college game produces good ones every year. Off-ball linebackers and safeties matter, but they win fewer one-on-one matchups that decide plays, and they are easier to find.

🔥 Common misconception

"Running backs don't matter." Positional value is about *price*, not about whether a position matters. Saquon Barkley ran for 2,005 yards for the 2024 Eagles and was central to a Super Bowl run (the Eagles case study tells it). The argument is narrower: the gap between a great back and a decent one is smaller than the gap between their salaries, so the money usually buys more at another position. The Eagles could pay Barkley because they had built cheaply elsewhere, which is the real lesson of positional value.

Does spending on a position buy wins?

If some positions are worth more, you might expect teams that spend more of their cap on them to win more. Figure 2 tests that directly, for every team-season from 2014 to 2025: how much of its player spending went to each position group, against its winning percentage.

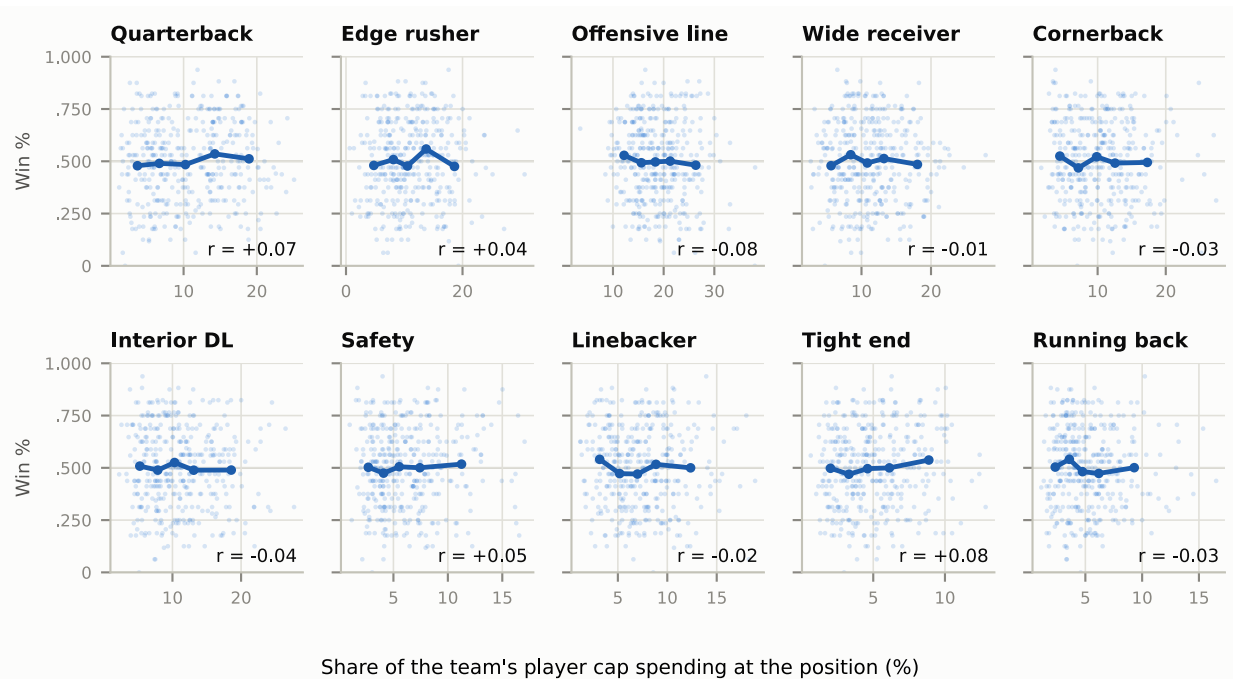


Figure 2: Positional cap share against winning, every team-season 2014–2025. Each faint dot is one team-season: the share of the team's player cap spending that went to that position group (x) and its regular-season winning percentage (y). The dark line joins the average winning percentage in each fifth of the spending range; r is the correlation. Every line is close to flat: how much a team spent on a position, quarterback included, says almost nothing about how often it won. Dead money is not included, so shares are of money spent on players on the roster. Data: Over the Cap contract histories via nflverse; nflverse schedules.

Every line is close to flat. Across 384 team-seasons, the correlation between a position group's share of spending and winning is never stronger than 0.08 in either direction, and for quarterbacks it is +0.07. Does that mean positional value is a myth? No. It means **price and value are different things**, for three reasons.

1. **The cap is zero-sum.** Spending 20 percent of the cap on a quarterback means spending 20 percent less on everyone else. A team that pays one star well and fills the rest with bargains can end up anywhere.
2. **Contracts pay for the past.** Teams sign big deals with players who *were* great, and pay them through their late twenties and early thirties, when many decline. A big share can be a reward for seasons already played.
3. **The best bargains are on rookie contracts.** The cheapest good players in the league are young ones on the rookie wage scale, which is the subject of the draft section. A team with a great young quarterback spends *less* on the position than a team overpaying a mediocre veteran.

The quarterback case makes the point clearest. Figure 3 sets the price of the position against the quality of the play.

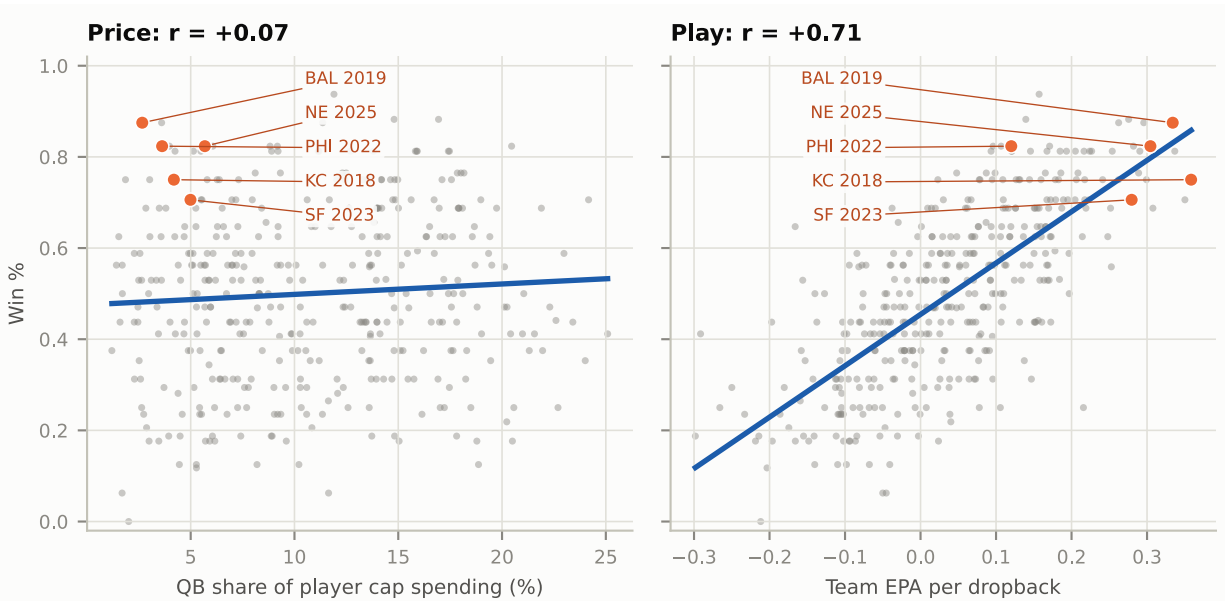


Figure 3: Price versus play at quarterback, every team-season 2014–2025. Left: the share of the team’s player cap spending that went to quarterbacks against winning percentage. It is close to flat. Right: the team’s EPA per dropback (every pass attempt, sack and scramble) against winning percentage. It is steep. The labelled seasons are teams whose starting quarterback was on a rookie contract (Mahomes in 2018, Jackson in 2019, Hurts in 2022, Purdy in 2023, Maye in 2025): elite play, low price, the most valuable combination in the sport. Data: Over the Cap via nflverse; nflverse play-by-play and schedules.

How well a team passes correlates with winning at 0.71, about as strong as anything in football statistics; how much it pays its quarterbacks correlates at 0.07. (EPA per dropback isn’t only the quarterback, of course: the line and receivers are in it too. But the quarterback is the largest single piece.) The value is in the play. The money is a separate question, and the gap between the two is where rosters are won. The labelled seasons show the best gap there is: a quarterback playing like a star on a rookie salary. Each of those teams used the savings to buy help; four of the five reached at least a conference championship game, and three reached the Super Bowl.

That is the whole logic of roster building in miniature. **Pay for value where you must, and get value cheaply wherever you can.** The tools for “cheaply” are the draft and the cap rules, and the next sections take them one at a time.

How a contract really hits the cap

A headline says “five years, \$150 million.” Almost none of that number tells you what the team will actually carry on its cap in a given year, or what it would cost to get out of the deal. Four ideas do.

The cap hit is what a contract counts against the cap in one league year: the player’s base salary for that year, plus the year’s share of his signing bonus, plus any roster, workout or incentive bonuses counted that year. It is an accounting number, not the cash paid. The broadcast graphic shows it because it’s the number that limits the team.

Signing-bonus proration is the rule that creates the gap between cash and cap. A signing bonus is paid to the player up front (often within months), but for cap purposes it is spread evenly over the years of the contract, up to a maximum of five. That is why teams love signing bonuses: the player gets the cash now, and the team spreads the cap cost over years when the cap will (usually) be higher.

Guaranteed money is money the team must pay even if it releases the player. It comes in flavors: *fully guaranteed at signing* (owed no matter what), *injury-only* (owed if he is cut while hurt), and money that **vests**, becoming fully guaranteed on a set date, often a few days into the new league year, if he is still on the roster. Agents and reporters often quote the total of all of these, which is why “total guarantees” in a headline can be far larger than what the player is truly certain to receive.

Dead money is the bill for leaving early. If a team releases or trades a player, every year of signing-bonus proration not yet counted lands on the cap at once, along with any guaranteed salary still owed. That money pays for nobody on the field.

Figure 4 walks one invented contract through all of it.



Figure 4: One invented contract, three ways. Five years, \$150 million: a \$50 million signing bonus and base salaries of \$5, 20, 25, 25 and 25 million. Top: as signed, the bonus counts \$10 million a year (light blue) on top of each year's base salary (dark blue); the cap hits are 15, 30, 35, 35 and 35. Middle: before year 3 the team restructures, turning \$20 million of that year's salary into a new bonus and spreading it over the three real years left plus two void years (aqua, \$4 million a year). Year 3's cap hit falls from 35 to 19, but years 4 and 5 rise, and when the void years cancel, their \$8 million lands as dead money (red). Bottom: the team cuts the player after year 2 instead. Before June 1, all \$30 million of unspent bonus becomes dead money in year 3, saving only \$5 million of cap; with a post-June 1 designation the dead money is split, \$10 million in year 3 and \$20 million in year 4.

Read the top panel first. The player is paid \$55 million in cash in year 1 (the \$50 million bonus plus \$5 million of salary), but the team counts only \$15 million against the cap. That gap is why every big contract is built around a large signing bonus: it front-loads the player's money and back-loads the team's cap cost. Typically the bonus plus the first year or two of salary are fully guaranteed at signing, here \$75 million, and the later years are not, which is what makes them "team-friendly": the team can walk away from them.

Restructures and void years: borrowing from the future

The middle panel is the most common move in the cap world, the **restructure**. A team needs cap space this year, so it asks a player it plans to keep to turn most of this year's base salary into a signing bonus. The player loses nothing (he gets the same money, sooner, and it becomes guaranteed); the team spreads the new bonus over the remaining years and cuts this year's cap hit sharply. The catch is that later cap hits rise by the same total, so a restructure is borrowing, not saving.

Void years stretch the loan. They are fake final years added to the contract that cancel automatically before anyone plays them. Their only purpose is to give the bonus more years to spread across, which lowers each year's charge. When they cancel, whatever proration was assigned to them becomes dead money all at once. Tom Brady's last Tampa Bay contracts were the famous example: after he retired in February 2023, the Buccaneers carried about \$35 million of dead money for a quarterback who was no longer playing.²

Cutting a player, and why June 1 matters

The bottom panel is the release. After year 2, the player's last three years carry \$30 million of proration that hasn't been counted. Cut him in March and all \$30 million is dead money this year, so the release saves only \$5 million of cap (his \$35 million cap hit minus \$30 million of dead money) even though it saves \$25 million of cash. That is why you hear "they can't afford to cut him": sometimes cutting a player costs nearly as much cap space as keeping him.

The **June 1 rule** softens it. If a player is released after June 1, or released earlier with a post-June 1 designation (each team gets two a year), only the current year's proration counts this year and the rest moves to next year. The team gets more room now and pays the remainder later.

i Film room: the Denver Broncos, March 2024

The move: Denver released quarterback Russell Wilson two years after trading for him and signing him to a five-year, \$245 million extension. The release created about **\$85 million of dead money**, the largest amount in league history at the time, and the Broncos used a post-June 1 designation to split it: about \$53 million on the 2024 cap and \$32 million on 2025.³

What to notice: why a team would accept that. Wilson's remaining guaranteed salary had to be paid either way, so keeping him cost the same money plus a roster spot for a player the coaches had benched. Dead money is a **sunk cost**: it is spent whatever you do next, and good front offices decide on what each choice costs *from here*. Denver spent 2024 with a rookie quarterback, Bo Nix, on a rookie contract, and made the playoffs; in 2025 it went 14–3 and earned the AFC's top seed while still carrying the second half of the Wilson charge. A painful cap year is survivable when the replacement is cheap.

Getting players: free agency, tags, trades and waivers

Every route onto a roster has a price, and the price is paid in different currencies: cap space, draft picks, or the right to a player in future years. Figure 5 puts the routes on the calendar, because when each one is open shapes how teams use it.

²Over the Cap's 2023 Tampa Bay Buccaneers cap table (dead money section) and contemporary reporting on Brady's retirement on February 1, 2023 (ESPN and NFL.com).

³ESPN and NFL.com reporting on Denver's release of Russell Wilson, March 4, 2024, and Over the Cap's Broncos dead-money tables for 2024 and 2025. Denver's 2025 record and seed: course fact sheet (FACTS-current §2).

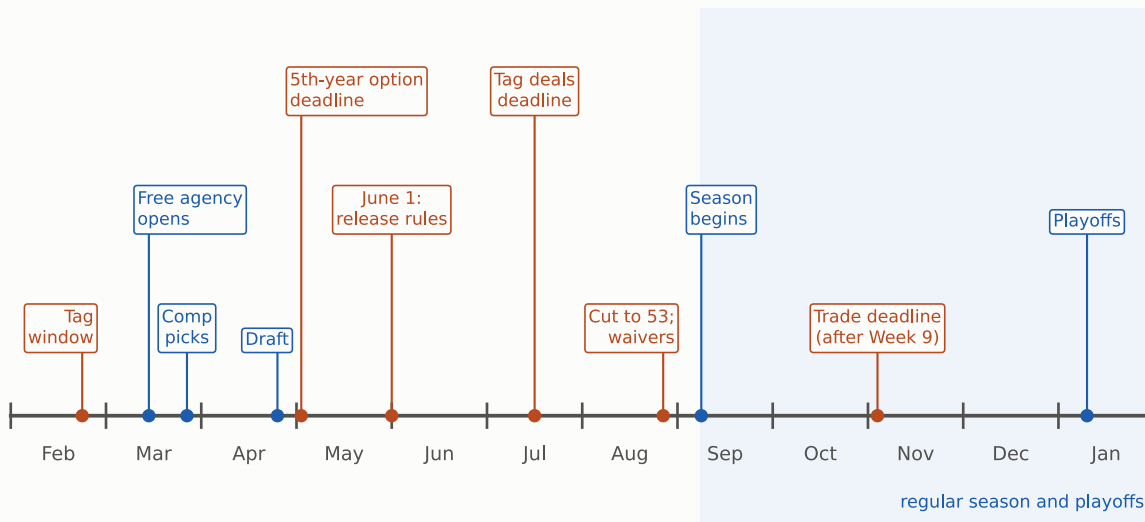


Figure 5: The NFL's business calendar (approximate dates; exact days shift each year). The league year turns over in mid-March, when contracts expire, free agency opens and trades become official. Tags are placed in the two weeks before, and a tagged player has until mid-July to sign a long-term deal; compensatory picks are awarded in March; the draft comes in late April and fifth-year options are due in early May; the June 1 date changes how releases count; rosters are cut to 53 in late August; the trade deadline falls in early November, after Week 9; and waivers run all year. Blue: acquiring players; orange: deadlines and cuts.

Free agency: unrestricted and restricted

When a contract expires, what happens depends on the player's **accrued seasons**, the seasons in which he spent at least six regular-season games on full pay status (on the active roster, inactive, or on injured reserve).

- An **unrestricted free agent (UFA)** has four or more accrued seasons. He may sign with anyone, and his old team gets nothing back except possibly a compensatory pick a year later (explained below). This is the market most fans picture: the first days of the league year in March, when the best players sign big deals, usually after a two-day negotiating window just before it.
- A **restricted free agent (RFA)** has exactly three. His team can offer him a one-year **tender** at one of several set prices. He may then talk to other teams and sign an **offer sheet**, but his old team can match it and keep him; if it doesn't, it receives a draft pick (first-round, second-round, or the round he was originally drafted in, depending on the tender) or, at the lowest tender, nothing. Offer sheets are rare, because teams don't like giving up picks to pry a player loose.
- A player with fewer than three is an **exclusive-rights free agent**: if his team offers the minimum, he can't negotiate with anyone else.

Why free agency exists: it is the players' half of the 1993 bargain that also created the cap (see 11-02). **What it costs teams:** free agents are usually paid for what they did on their last team, at the peak of their market, and a bidding war means the team that "wins" is the one that valued him most, often too much. Analysts call that the **winner's curse**, and it's why patient teams treat free agency as a way to fill specific holes rather than to build a roster.

The franchise and transition tags

What if the player is a star and the team can't agree on a long-term deal? It can **tag** him.

The **franchise tag** is a one-year contract that a team may impose on one expiring player a year. Its price is set by formula: the average of the top cap numbers at his position (the top five, measured against the cap over recent years), or 120 percent of his previous salary, whichever is higher. It comes in two forms. The **non-exclusive** tag lets him negotiate with other teams, but his team may match any offer, and if it doesn't it receives two first-round picks, a price that almost always ends the conversation. The **exclusive** tag, at a slightly higher price, forbids him from negotiating with anyone else at all.

The **transition tag** is the cheaper version, priced from the top ten salaries at the position. Other teams can sign the player to an offer sheet, and his team gets only the right to match, with no compensation if it lets him go.⁴

⁴The franchise and transition tags are defined in Article 10 of the 2020 NFL-NFLPA Collective Bargaining Agreement, via Over the Cap: <https://overthecap.com/collective-bargaining-agreement/article/10>. Restricted and unrestricted free agency are in Article 9, <https://overthecap.com/collective-bargaining-agreement/article/9>.

Why tags exist: to let a team keep a great player for one more season while the two sides keep negotiating; a tagged player and his team have until mid-July to agree on a long-term deal, or he plays the year on the tag. **What a tag costs:** a full year of top-of-market salary, fully guaranteed, with no cap-friendly bonus structure, and a player who usually feels he has been denied his long-term security. It also gets expensive fast: a second tag of the same player costs 120 percent of the first, and a third costs 144 percent of the second. Washington tagged quarterback Kirk Cousins in 2016 and again in 2017, then let him walk; Minnesota signed him in 2018 to a three-year contract that was **fully guaranteed**, a rarity at the time and a direct result of the leverage two tags had given him.

Compensatory picks

A team that loses more good free agents than it signs gets a consolation prize. The **compensatory pick** formula looks at every unrestricted free agent who left or arrived the previous year, ranks each by his new contract's average salary, playing time and honors, and awards up to 32 extra picks league-wide, at the end of rounds 3 through 7, at most four per team. Since 2017 the picks can be traded. A separate rule, adopted in 2020, awards third-round picks in two straight drafts to a team that loses a minority assistant coach or executive to another team's head-coach or general-manager job.⁵

Why it matters for strategy: a team can *choose* to let its own free agents leave and avoid signing others' in order to collect picks. The Baltimore Ravens are the model: they have received more compensatory picks than any other franchise since the system began in 1994, and their roster is built on letting expensive veterans walk at the right time.

Trades and the trade deadline

A trade moves a contract, not just a player. The acquiring team takes over his remaining salary and cap hits; the trading team keeps any signing-bonus proration it has not yet counted, as dead money. That is why trades cluster around the new league year (when teams reset their caps) and why a team "eats" money to make a deal work.

The **trade deadline** is the last day of the season on which trades are allowed: since 2024, the Tuesday after Week 9. After it, a contender's only additions are free agents nobody signed, waiver claims and its own practice squad. Contenders use the deadline to **rent** a veteran for a playoff run, often paying with picks in a future draft; sellers use it to turn a player who won't be part of their next good team into a pick that might be.

Waivers

When a team releases a player with fewer than four accrued seasons (and, after the trade deadline, any player), he goes on **waivers**: for about a day, every other team may claim him and take over his contract as it stands. If several teams claim him, the one with the worst record wins. (Until the fourth week of the season, priority follows the previous season's standings, the same order as the draft.) A player nobody claims becomes a free agent and can choose his next team. Waivers matter most on the day rosters are cut to 53 in late August, when hundreds of players are released at once and every front office combs the list for someone better than its own 53rd man.

⁵Compensatory picks: NFL Football Operations, "Compensatory Draft Picks" (operations.nfl.com, under the draft section), and the 2020 CBA's free-agency article cited above. The 2020 resolution on picks for teams that develop minority coaches and executives: NFL.com reporting, November 10, 2020.

The build: the Rams of 2021 are the boldest trade-built champion of the cap era. In March they made official the trade for quarterback Matthew Stafford, sending Detroit Jared Goff, a 2021 third-round pick and their first-round picks in 2022 and 2023. They already had cornerback Jalen Ramsey, acquired in 2019 for two first-round picks and a fourth. On November 2, at the trade deadline, they added edge rusher Von Miller for a 2022 second- and third-round pick. A week later Odell Beckham Jr., released by Cleveland, cleared waivers, which made him a free agent free to pick his team, and he chose Los Angeles.⁶ On February 13, 2022, the Rams beat Cincinnati 23–20 in Super Bowl LVI, with Stafford, Miller and Beckham all central to the win.

What to notice: the currency. The Rams paid in *future* draft picks, which a team with a short window values least, and spent them on premium positions (a quarterback, a cornerback, an edge rusher) rather than on depth. General manager Les Snead summed up the philosophy on a T-shirt at the victory parade with a phrase that can't be printed here, the gist being that picks were made to be traded. The bill came later: the Rams did not make a first-round pick in any draft from 2017 through 2023 (their first after Jared Goff in 2016 was Jared Verse, 19th overall in 2024), and went 5–12 in 2022 with an injured, expensive roster. They drafted their way back (Puka Nacua was a 2023 fifth-round pick) and were back in the playoffs in 2023. A trade-built roster wins a window; it doesn't build a pipeline.

The draft: the cheapest good players in football

The draft is where the gap between value and price is widest, because of one rule.

The rookie wage scale and the fifth-year option

Before 2011, top draft picks negotiated like free agents, and the price had run away: the 2010 first overall pick, quarterback Sam Bradford, signed a six-year deal worth up to \$78 million with \$50 million guaranteed before playing a down, more than many proven veteran quarterbacks earned. The 2011 collective bargaining agreement ended that with the **rookie wage scale**. Every draft slot now carries a set four-year contract value, falling steeply from the first pick to the last, with little left to negotiate. Drafted players cannot sign an extension until after their third regular season.⁷

First-round picks come with one extra year of team control. The **fifth-year option** lets the team add a fifth season at a set price, decided after the player's third season (by early May). Since the 2020 agreement the option is fully guaranteed once exercised, and its price depends on how much the player has played and whether he made Pro Bowls, so teams now decline it for players they aren't sure of.

Together these rules create the **rookie-contract window**: three to five years in which a good young player is paid far less than a veteran of his quality. For most positions the window is a nice bonus. For a quarterback it is the most valuable asset in the sport, as Figure 3 showed, and teams with a young starter build their entire cap plan around it.

Draft value charts: what is a pick worth?

To trade picks, teams need a common price list. The original **draft value chart** was built for the Dallas Cowboys in the early 1990s, under head coach Jimmy Johnson, from the prices of trades teams had actually made; Cowboys part-owner Mike McCoy is usually credited with the arithmetic. The **Jimmy Johnson chart** gives the first pick 3,000 points, the 16th 1,000, the 32nd 590, the 64th 270 and the 100th 100. Teams still use it or close cousins, and a trade that "balances on the chart" is still how deals get done.

But the chart describes what teams *pay* for picks, not what the picks *produce*. In 2005 two behavioral economists, Cade Massey and Richard Thaler (Thaler later won a Nobel prize for work on how people make predictable mistakes), asked whether the prices were right. They measured each pick's **surplus value**: what a player produces minus what he is paid. Their answer, published as "The Loser's Curse," was that teams badly overvalue the top of the draft. A top pick is more likely to become a star, but he is also paid far more, and teams are overconfident in their ability to tell the stars apart. By surplus value, they found, picks *gained* value

⁶Trade details from nflverse's trade table (compiled from Pro Football Reference), loaded in this chapter's code with `load_trades()`: Stafford (official March 18, 2021; Goff, 2021 No. 101, 2022 No. 32 and 2023 No. 6 to Detroit), Miller (Nov. 2, 2021: 2022 Nos. 64 and 96 to Denver), Ramsey (Oct. 16, 2019: 2020 No. 20, 2021 No. 25 and 2021 No. 130 to Jacksonville). Rams first-round picks from nflverse draft data. Beckham's release by Cleveland and signing with the Rams: NFL.com reporting, November 2021.

⁷Bradford's contract and the motivation for the scale: ESPN reporting on his July 2010 contract with the Rams. The rookie compensation system and the fifth-year option are Article 7 of the 2020 CBA, <https://overthecap.com/collective-bargaining-agreement/article/7/>; the option's guarantee and tiered price were changes in the 2020 agreement.

through the first round, and the best value in the draft sat around the early second round.⁸ The curse of the title: the worst team earns the first pick, and the first pick is overpriced.

Their data came from before the rookie wage scale, when top picks were far more expensive. We can redo the idea for the current era with contract data. For every player drafted from 2011 to 2020, take what the market paid him on his **next** contract (as a share of the cap) as the measure of what he turned out to be worth, and take his rookie salary as what he cost. Surplus is the difference. Figure 6 compares that curve with the Johnson chart.

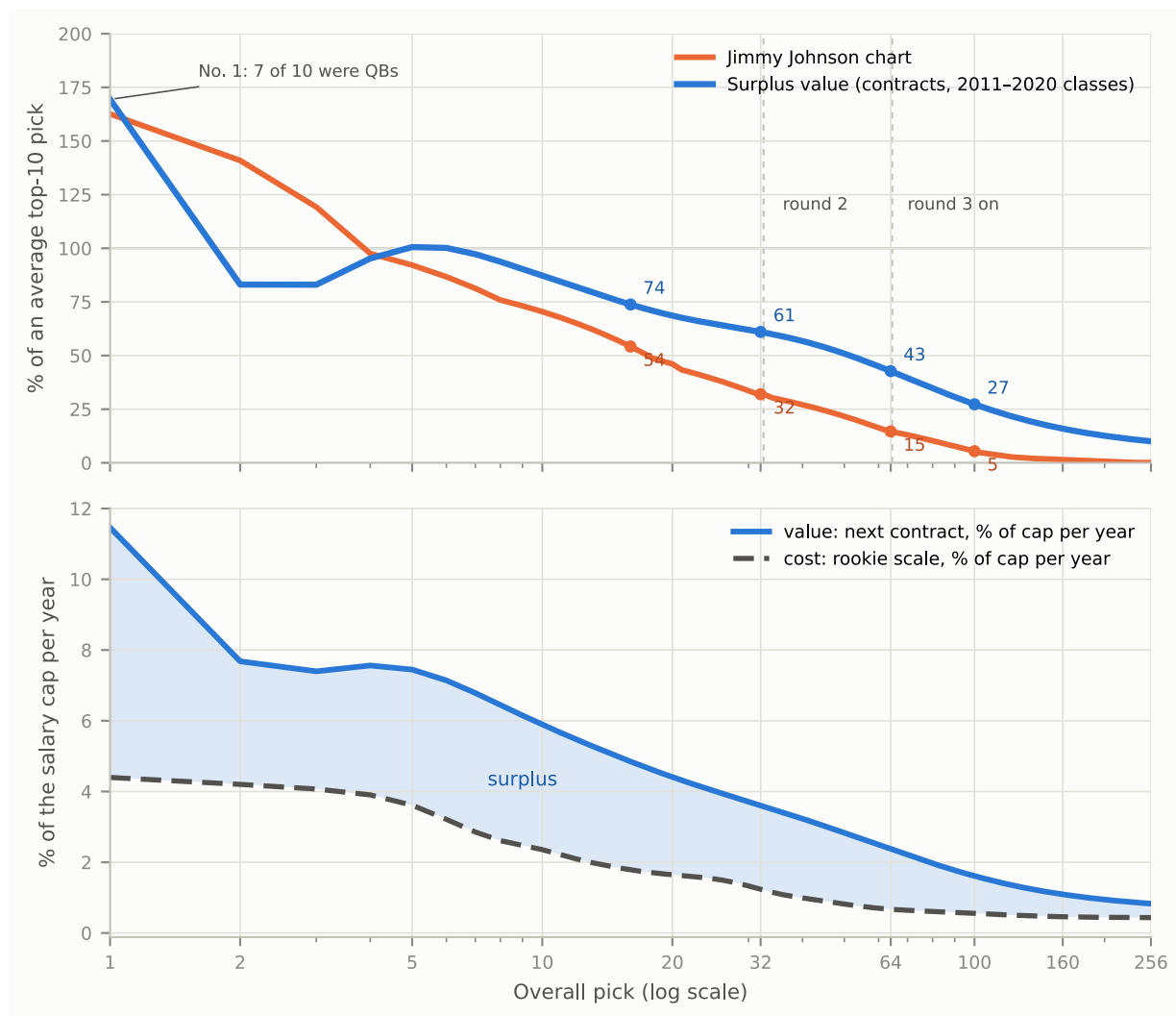


Figure 6: Two price lists for the draft. Top: each pick's value as a percentage of an average top-ten pick, by the Jimmy Johnson chart (orange) and by surplus value (blue, smoothed). They agree that the first pick is special: in the 2011–2020 drafts it was usually a quarterback. After that they part ways. The Johnson chart says the 32nd pick is worth about a third of a top-ten pick; by surplus it is worth about 60 percent, and the 64th pick about 40 percent rather than 15. Bottom: where the surplus comes from, in percent of the salary cap per season: the rookie wage scale (cost, 2021–2025 contracts) falls far faster than what those players went on to earn on their next contract (value, 2011–2020 draft classes; zero for a player who never signed one). The shaded gap is surplus.

Both panels use a log scale for the pick, because value changes fastest at the top. Data: Over the Cap contract data via nflverse.

The two curves agree on the order (an earlier pick is always worth more) and on the first pick, and disagree wildly about everything after it. Measured against an average top-ten pick, the Johnson chart prices the 32nd pick at 32 percent and the 64th at 15; by surplus they keep about 61 and 43 percent. The bottom panel shows why: the rookie scale makes the first pick cost about 4.4% of the cap a year, about 7 times what a third-round pick costs, but what the players turn out to be worth falls far more gently. The modern data softens Massey and Thaler's "curse" (the wage scale made the top picks cheaper, so the very top is no longer a bad

⁸Cade Massey and Richard H. Thaler, "The Loser's Curse: Decision Making and Market Efficiency in the National Football League Draft," *Management Science* 59, no. 7 (2013): 1479–1495. A working-paper version circulated from 2005. The modern curve in this chapter uses a different measure of value (the next contract, an approach similar to one popularized by the analyst Ben Baldwin) and only post-2011 contracts, so its numbers differ from theirs.

deal), but their main finding survives: **the Johnson chart drops far too fast after the first few picks**, so a team trading *down* for more picks is usually buying more value than the chart says, and a team trading *up* is usually paying more.

Two caveats keep this honest. First, “next contract” is a rough measure of value: it misses players who were excellent and then hurt, and it credits players paid on reputation. Second, the top of the curve is built from very few players (ten per pick over ten drafts), and it is dominated by quarterbacks: 7 of the ten first overall picks from 2011 to 2020 were quarterbacks. That is the one place where the old chart’s steepness can be right. A franchise quarterback is worth so much more than anything else that a team convinced it has found one can rationally pay almost any price, which is why the biggest trade-ups are for quarterbacks. The problem is that the conviction is often wrong, as the next film room shows.

i Film room: Brock Purdy and the 49ers, 2021–2025

The setup: In March 2021 the 49ers traded up from the 12th pick to the 3rd, giving Miami the 12th pick, their first-round picks in 2022 and 2023, and a 2022 third-rounder, and used it on quarterback Trey Lance. Lance started four games in two seasons, broke his ankle in 2022, and was traded to Dallas in August 2023 for a fourth-round pick. The quarterback who took his job was Brock Purdy, the 262nd and last pick of the 2022 draft, whose four-year rookie contract was worth \$3.7 million in total.⁹

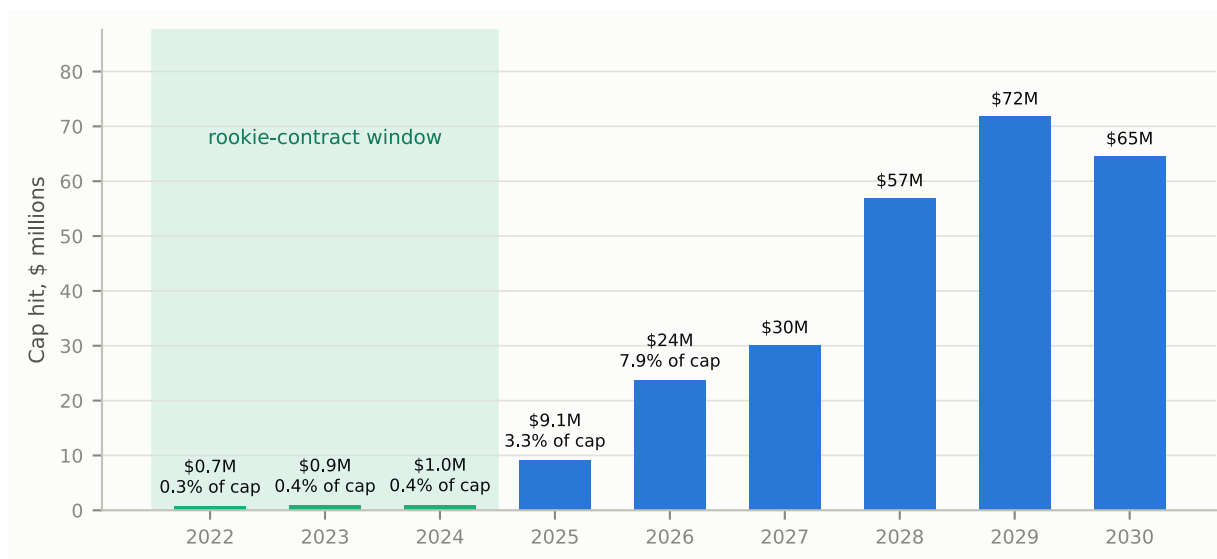


Figure 7: Brock Purdy’s cap hits, 2022–2030, in \$ millions, with each year’s share of that year’s salary cap above the bar where the cap is known. The three rookie-contract seasons (shaded) cost the 49ers less than half a percent of the cap each. The five-year extension signed in May 2025 is shaped like most big contracts: a modest first year, then rising hits as the cap grows.

Years after 2026 are as scheduled at signing and will change if the deal is restructured. Data: Over the Cap via nflverse.

What to notice: what the window bought. In 2023, with Purdy’s cap hit under a million dollars, the 49ers could pay Christian McCaffrey, Nick Bosa, Trent Williams, Fred Warner, George Kittle and Deebo Samuel at the same time, and they reached the Super Bowl. In May 2025 the window closed: a five-year extension worth up to \$265 million, \$53 million a year, about 19% of the cap when signed. The same team, in the same two drafts, paid three first-round picks for the quarterback the Johnson chart loved and got its starter for a seventh-round pick no chart valued at all. That is not a lesson that late picks beat early ones; it is a lesson in how uncertain the draft is, and why surplus value says to collect more chances rather than pay heavily for one.

Evaluating players: what scouts grade, and what the stopwatch can’t see

Every pick starts with an evaluation, and the core of evaluation is still film: area scouts watch every snap a prospect played in college and grade him against what his position demands. Figure 8 is a condensed version of the traits most NFL scouting departments grade.

⁹Trade and contract data from nflverse (`load_trades()`, `load_contracts()`, Over the Cap). Lance was traded to Dallas on Aug. 25, 2023, for a 2024 fourth-round pick. Purdy was the 262nd pick in 2022.

Quarterback GRADED ON FILM - Processing: reads the coverage, finds the open man on time - Accuracy and ball placement, not just completion % - Arm: velocity and every throw to the far sideline STAND-IN NUMBERS hand size; little else THE STOPWATCH MISSES poise when the pocket collapses	Offensive tackle GRADED ON FILM - Length: arms 34 in. or more to keep rushers off his body - Anchor: absorbs a bull rush without giving ground - Feet: kick-slide to mirror speed rushers STAND-IN NUMBERS arm length, 10-yd split, short shuttle THE STOPWATCH MISSES hand timing and recovery
Edge rusher GRADED ON FILM - Get-off: first-step burst at the snap - Bend: dips under the tackle's reach at full speed - Hands and a plan: moves and counters STAND-IN NUMBERS 10-yard split, 3-cone, length THE STOPWATCH MISSES motor in the fourth quarter	Interior DL GRADED ON FILM - Quickness to shoot a gap (one-gap) or strength to control a blocker (two-gap) - Leverage: plays low - Hands to shed blocks STAND-IN NUMBERS weight, 10-yd split, bench press THE STOPWATCH MISSES playing low when tired
Wide receiver GRADED ON FILM - Release against press - Route craft: tempo, breaks, how he sets up a defender - Hands and catch radius; run after the catch STAND-IN NUMBERS 40-yard dash, height, vertical jump THE STOPWATCH MISSES separation against NFL corners	Cornerback GRADED ON FILM - Press and mirror: stays on the receiver's hip off the line - Long speed: can run with a go route for 40 yards - Ball skills: eyes in zone STAND-IN NUMBERS 40-yard dash, height, arm length, 3-cone THE STOPWATCH MISSES recovering after being beaten

Figure 8: What scouts grade, position by position (a composite of common scouting criteria, not one team's form). For each position: the traits that decide the grade, the measurement or drill that stands in for them, and what no stopwatch can see. Notice how much of each card is about processing, technique and movement in pads rather than raw speed: that is why the combine adjusts grades rather than sets them.

A few words on those cards are terms of art. A lineman's **anchor** is his ability to stop a bull rush, sinking his weight and holding his ground instead of being driven back into the quarterback (Pass Protection). An edge rusher's **get-off** is his first-step explosion at the snap, and **bend** is his ability to dip his shoulder and turn the corner around a tackle at full speed without slowing down (The Pass Rush). **Processing**, for a quarterback, is how quickly he reads the coverage and knows where to throw. Notice that the first trait on almost every card is one no stopwatch measures.

The combine and pro days

Film is graded through the fall. In February and March come the measurements. The **NFL Scouting Combine** brings roughly 300 invited prospects to Indianapolis, where it has been held since 1987, for a week of medical exams, interviews with teams, body measurements (height, weight, arm length, hand size) and timed drills.¹⁰ Each drill stands in for a football trait:

¹⁰NFL, "NFL Scouting Combine" overview, <https://www.nfl.com/combine/>. The combine's move to Indianapolis in 1987 is from the league's and the host city's published histories.

- **40-yard dash**, with a **10-yard split**: long speed for receivers, backs and corners; the split, the first ten yards, is the closest number to a lineman's or edge rusher's burst off the snap.
- **Vertical and broad jump**: lower-body explosiveness.
- **Bench press** (reps of 225 pounds): upper-body strength endurance, mostly for linemen.
- **Three-cone drill** and **20-yard short shuttle**: change of direction, the ability to stop, bend and restart, which matters for edge rushers turning the corner and corners mirroring routes.

A **pro day** is the campus version, held in March and April for a school's prospects with NFL scouts and coaches watching. Players who skipped drills at the combine run them here, usually on their own familiar turf and often with friendlier timing, so experienced evaluators treat pro-day numbers as slightly optimistic.

RAS: putting the numbers on one scale

A 4.6 forty is slow for a cornerback and astonishing for a 300-pound tackle, so raw numbers can't be compared across positions. **RAS**, the **Relative Athletic Score** created by the analyst Kent Lee Platte, solves that by converting each measurement into a percentile among every player measured at the same position since 1987, then combining them into one score from 0 to 10. A 5 is average for the position; 8 is very good; 9 and above is elite.¹¹

Is it useful? Figure 9 builds a simplified RAS-style score from the nflverse combine data and asks what it predicts.

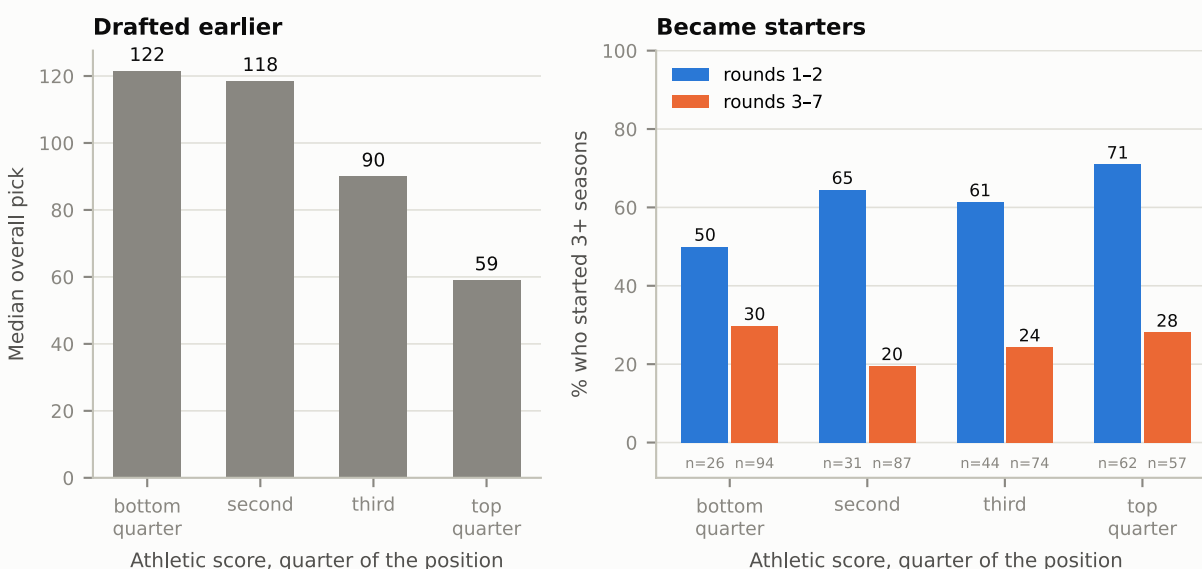


Figure 9: What an athletic score predicts, and what it doesn't. A simplified RAS-style score (each of height, weight, 40-yard dash, vertical, broad jump, bench, three-cone and shuttle converted to a percentile within the position, averaged and scaled 0–10; at least five measurements required) for edge rushers and cornerbacks at the 2006–2018 combines who were drafted. Left: better athletes are drafted earlier: teams already price athleticism in. Both panels split the players into quarters by their score within the position. Right: the share who became starters for three or more seasons, split by when they were drafted. Within a draft-round group the bars change only modestly. Once teams have used the numbers to set the draft order, a high score adds a little, not a lot. Data: nflverse combine and draft-pick data (Pro Football Reference).

The left panel shows that teams already draft by athleticism: the correlation between the score and the pick number is -0.34 (negative because a better score means an earlier, lower-numbered pick). The right panel shows the limit. Athletic scores tell you who *can* play the position at NFL speed; they can't tell you who processes a coverage, plays with his hands, or keeps improving. So treat **RAS as a filter, not a ranking**: a very low score at a premium position is a real warning, and a very high score on a player with thin film is a bet on coaching. Some players beat their numbers for a career. Others, the “workout warriors,” turn a great combine into a high pick and never play like it.

¹¹Kent Lee Platte, “What is RAS?”, <https://ras.football/whats-ras/>. This chapter's score is a simplified imitation (equal weights, nflverse's combine measurements only), not RAS itself.

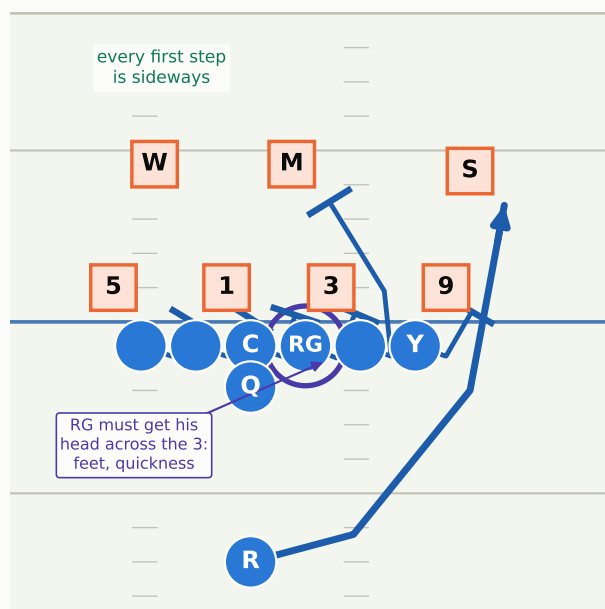
Scheme fit: the same player, different value

Here is where X's and O's come back. **Scheme fit** is how well a player's body and skills match what a particular scheme asks of his position. Because schemes ask different things, the same prospect can be a first-round talent on one board and a fourth-rounder on another, and a smart team shops where its scheme makes it the only serious buyer.

Offensive line: move your feet or move the man

A wide-zone team and a gap team ask their linemen to do opposite things. Outside zone (wide zone) asks every lineman to step sideways at the snap and reach the defender in front of him: to get his body across a man who is already lined up outside him. That is a race, and it rewards quick feet, balance and athleticism over size. A gap scheme such as power asks the play-side linemen to down block, driving a defender sideways off his spot from an angle, and asks a guard to pull and lead through the hole. That rewards mass and power, plus one guard who can run. Figure 10 puts the two side by side.

Outside zone: win the race



Power: move the man

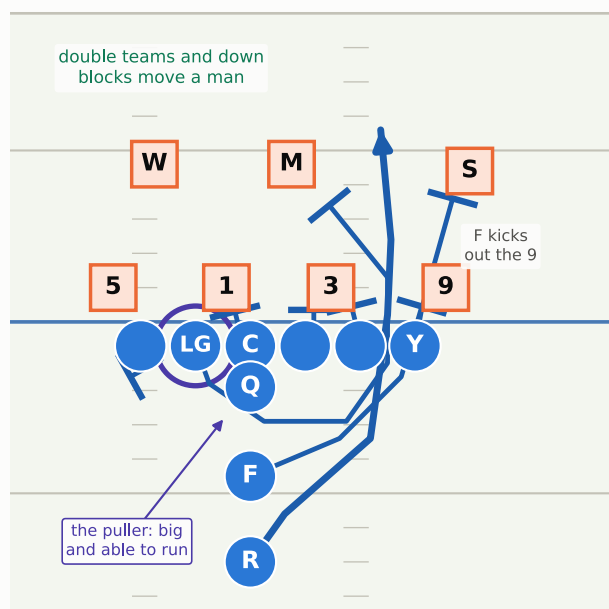


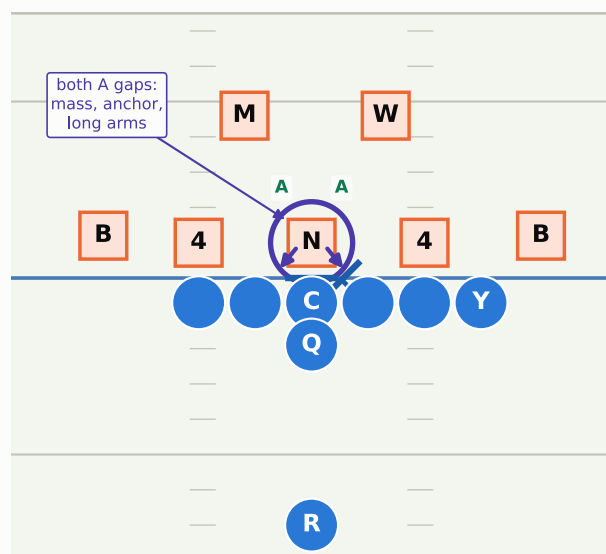
Figure 10: Why a wide-zone team and a gap team draft different linemen (illustrative, against the same four-man front). Left, outside zone right: every blocker's first step is sideways, and the right guard (ringed) has to get his body across a 3-technique who starts outside him. That is a race won with feet and quickness. Right, power right: the right guard and tackle double-team the 3-technique and drive him off the ball, the center blocks back on the 1, the fullback kicks out the end, and the left guard (ringed) pulls around to lead through the hole for the Mike. Those blocks are won with mass and leg drive, and the puller is a big man who must also run. Typical builds: wide-zone linemen often weigh about 295–315 pounds, gap linemen 315–340.

That's why a wide-zone team will happily draft a 300-pound guard who ran well in the shuttle and a gap team will pass on him for a 330-pounder who moves people. The weights are typical rather than fixed: wide-zone linemen are often in the 295–315 range and gap linemen 315–340, but teams draft for what the player does on film, and a heavy man who moves well fits anywhere.

Defensive line: two-gap nose or one-gap 3-technique

The same split runs through the defensive front. In a two-gap defense, often a 3-4 with a nose tackle head-up on the center, the lineman's job is to strike the blocker, control him, and be ready to shed into *either* gap beside him (two-gap play). In a one-gap attacking front, the 3-technique on the guard's outside shoulder owns a single gap and shoots into it at the snap (one-gap play). DL Techniques and Gap Control taught both jobs; Figure 11 shows what each asks of a body.

Two-gap nose: absorb and control



One-gap 3-technique: penetrate

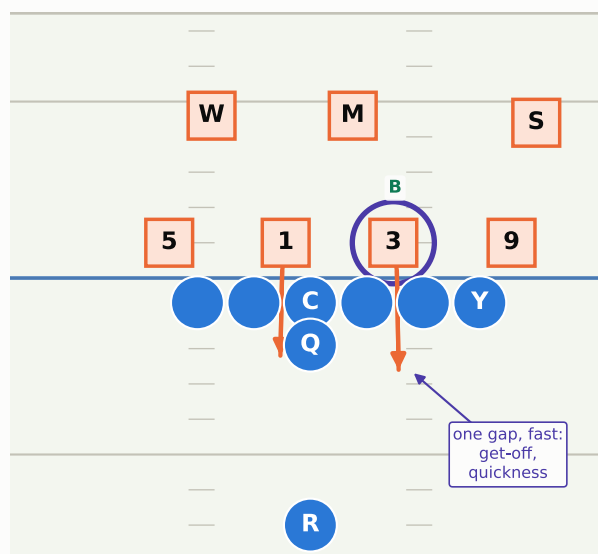


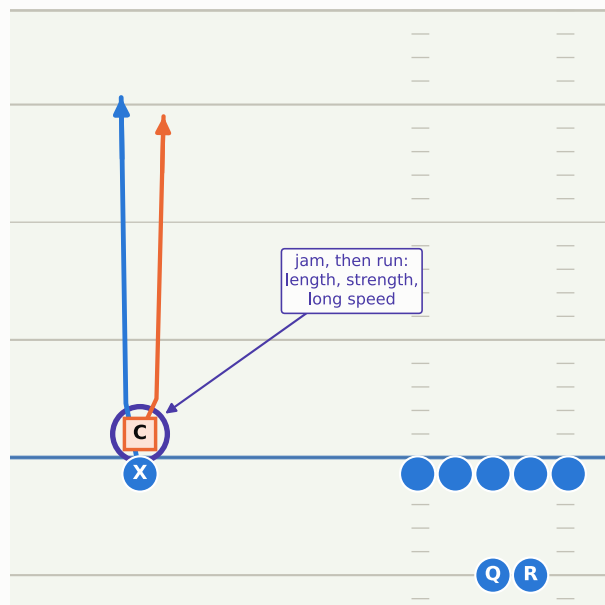
Figure 11: Two defensive-line jobs, two body types (illustrative). Left, a two-gap 3-4: the nose (ringed) lines up head-up on the center and must absorb him, often a guard as well, and still be able to shed into either A gap. He needs mass, a strong anchor and long arms, so classic two-gap noses are often 320–340 pounds or more. Right, a one-gap four-man front: the 3-technique (ringed) is shaded on the right guard's outside shoulder and attacks the B gap at the snap. He needs first-step quickness and leverage more than bulk, so he is often 285–305 pounds. A team switching from one front to the other has to rebuild its defensive line.

The edge has the same split. A 4-3 defensive end plays with his hand in the dirt and must hold up against tackles in the run game, so teams like him heavier, around 265–285 pounds; a 3-4 outside linebacker stands up, rushes from wider and sometimes drops into coverage, so he can be lighter and quicker, around 240–260.

Cornerbacks: press or off

Cornerbacks split by technique. A team that plays a lot of man coverage with its corners in press, jamming the receiver at the line, wants long, physical corners with the hips to turn and the long speed to run with a go route: the Seattle defenses of the early 2010s famously preferred corners at least six feet tall with long arms. A team that plays more zone and off coverage, its corners seven yards off with eyes on the quarterback, can live with shorter corners who read the play well and break on the ball quickly. Figure 12 shows both alignments.

Press-man: stay on his hip



Off zone: eyes on the QB

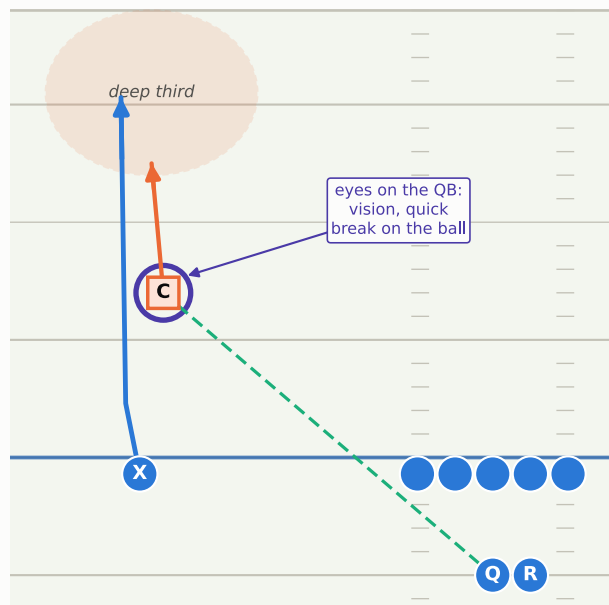


Figure 12: Two cornerback jobs (illustrative; the outside receiver on the offense's left). Left, press-man: the corner (ringed) is a yard off the receiver, jams him, then turns and runs with him on his hip, no matter where he goes. Length, strength and long speed decide it. Right, off zone (a Cover 3 deep third): the corner is seven yards off, backpedals with his eyes on the quarterback, and drives on any throw in front of him. Vision, footwork and the quick break on the ball decide it, so a shorter corner can thrive.

Scheme fit reaches all the way to your fantasy team, too. A receiver who runs 600 routes a season in one offense might run 400 in another, and a running back who catches passes in one scheme only carries the ball in the next, which is why the same player's fantasy value can jump or collapse when he changes teams. That idea, reading a player's *role* in the scheme rather than his box score, is the subject of Scheme and Fantasy.

The archetypes, in real numbers

Figure 13 puts real combine measurements behind these archetypes. Each dot is one prospect; the shaded regions are the typical scheme archetypes, drawn by hand to show where each kind of team shops.

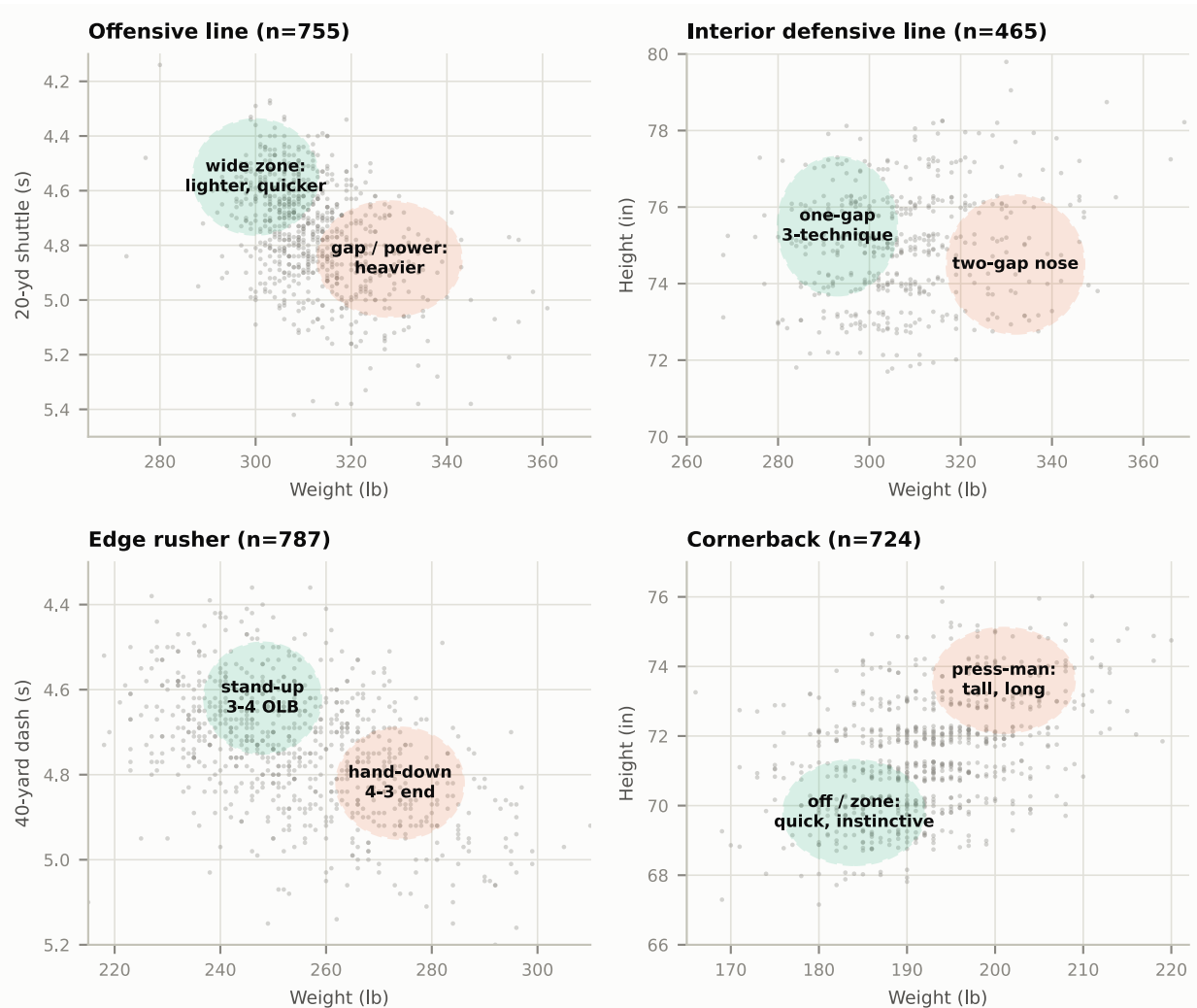


Figure 13: Body types by position and scheme: every combine prospect measured 2006–2026 at four positions (dots), with the typical scheme archetypes shaded (illustrative regions, not hard cut-offs). Offensive linemen: wide-zone teams shop on the lighter, quicker side (left, low shuttle times), gap teams on the heavier side. Interior defensive linemen: two-gap noses are the heaviest men in the draft; one-gap 3-techniques are 30 to 50 pounds lighter. Edge rushers: hand-down 4-3 ends are heavier than stand-up 3-4 outside linebackers. Cornerbacks: press-man teams want height (and length, which the combine measures but this data set does not include); off-coverage teams can take shorter players. Data: nflverse combine data (Pro Football Reference).

i Film room: the 49ers' mid-round wide-zone linemen, 2017–2025

The pattern: Kyle Shanahan's 49ers run the purest wide-zone offense in the league (the Shanahan case study tells its history), and they have mostly filled their offensive line from the middle of the draft. From 2017 to 2025 they drafted ten offensive linemen; only two went in the first two rounds (Mike McGlinchey ninth overall in 2018 and Aaron Banks in the second round in 2021), and the rest came from rounds 3 to 7: Dominick Puni (third round, 2024), who started as a rookie, Spencer Burford (fourth, 2022), Colton McKivitz (fifth, 2020) and others.¹²

What to notice: it isn't that the 49ers draft small linemen. Their picks with combine weights average about 311 pounds, close to the league's 314 for drafted linemen. What they shop for is movement (the short shuttle and the ability to reach and climb on film) at mid-round prices, trusting their coaching to turn those players into starters. And where positional value says to pay, they paid: in 2020 they traded for left tackle Trent Williams rather than wait for one to develop. Cheap, scheme-specific depth inside; a premium price at the premium position. That is positional value and scheme fit working together.

¹²nflverse draft picks (Pro Football Reference) for 2017–2025; combine weights from nflverse's combine data. The 49ers acquired Trent Williams from Washington in April 2020 for a 2020 fifth-round and 2021 third-round pick.

i Film room: the Eagles' offensive line

The Eagles took the opposite route to an elite line: they *paid* for it. Using the same contract data as Figure 2, Philadelphia's offensive line took a larger share of its player spending than the league average in every season from 2014 to 2025, as Figure 14 shows. But much of the talent was developed rather than bought: Jason Kelce was a sixth-round pick and Jordan Mailata, an Australian rugby league player who had never played American football, a seventh-rounder in 2018. The Eagles case study tells how that line made the scheme possible.

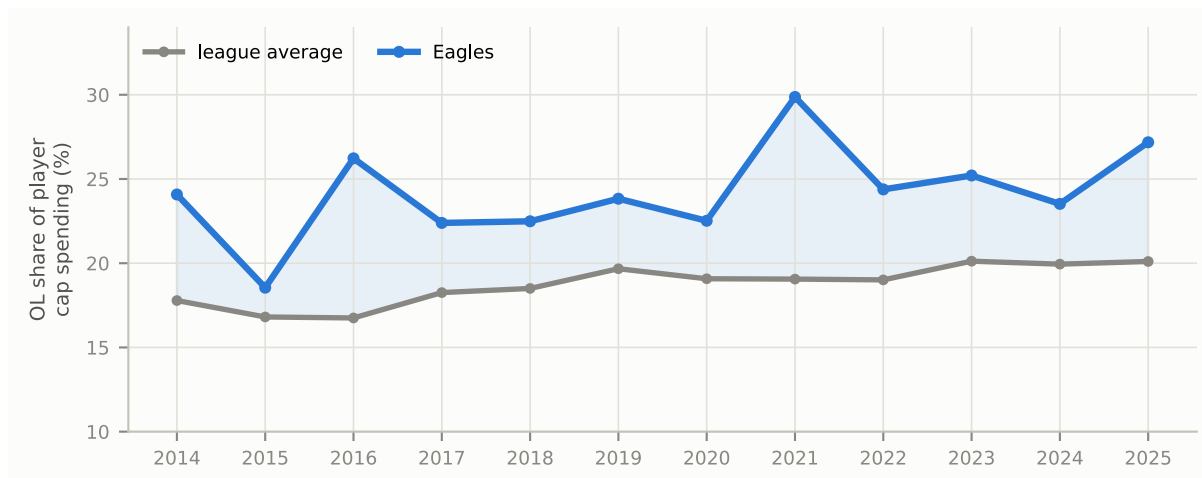


Figure 14: The Eagles' offensive line as a share of the team's player cap spending (blue) against the league average (grey), 2014–2025. Philadelphia was above the league in every season, and in half of them by five percentage points or more, roughly an extra starter's salary committed to the line each year. Dead money is excluded. Data: Over the Cap contract histories via nflverse.

The 53, the 48 and the practice squad

The last layer of roster building is the rulebook for who may play on Sunday. A team carries up to 90 players in the spring and summer, cuts to 53 in late August, and then manages several lists at once. Figure 15 lays them out.



Figure 15: How an NFL roster is organized during the season (rules as of 2026). The 53-man roster is the pool; on game day up to 48 of them may be active (if at least eight are offensive linemen), with the rest inactive. A third quarterback from the 53 may also dress as the emergency third quarterback without counting toward the 48. The practice squad (16 players, plus an international player exemption) practices with the team and can be elevated for a game, two a game and up to three times per player per season. Injured reserve removes a player from the 53; he must miss at least four games, and a team may bring back up to eight players a season.

Game-day actives. A team may dress 48 of its 53, provided at least eight are offensive linemen (otherwise 47); the rest are declared inactive about 90 minutes before kickoff. Who sits is a weekly game-plan decision. A team facing a run-heavy opponent might dress a fifth defensive tackle instead of a sixth receiver, and the inactive list on Sunday morning is one of the best free clues to the plan.

The practice squad. Each team keeps 16 players on its practice squad (plus one exemption for an international player), and up to six of them may be veterans with any amount of experience. They practice with the team but are not on the 53, and any other team may sign one to its 53 at any time. A **practice squad elevation** moves one of them up for a single game: a team may elevate two a game, each player at most three times a season, and he reverts to the practice squad afterward without going through waivers. That is how a team covers a Saturday injury without cutting anyone, and it's why a "practice squad" kicker can make a game-winning field goal.¹³

Injured reserve (IR) takes an injured player off the 53 so his spot can be filled. During the season a player placed on IR must sit out at least four games, and a team can bring back a limited number of players a year (eight); everyone else stays out for the season. The rule creates a real decision: put a player with a six-week injury on IR, freeing his spot for a healthy body, or keep him on the 53 and play a man short for a month and a half.

The emergency third quarterback. Since 2023 a team may dress a third quarterback from its 53 without counting him among the 48, and he may enter only if both of the other quarterbacks are injured or ejected. The rule was a direct answer to the 2022 NFC Championship Game, in which the 49ers lost Brock Purdy and then his backup Josh Johnson to injuries and finished the game with running back Christian McCaffrey taking snaps.¹⁴

¹³Practice squad, elevation and injured reserve rules: NFL Football Operations roster rules (operations.nfl.com) and NFL.com reporting on the 2022 changes (16-player squads, two elevations a game, three per player, eight IR returns).

¹⁴CBS Sports, "Emergency third quarterback rule explained" (2023), <https://www.cbssports.com/nfl/news/emergency-third-quarterback-rule-explained-things-to-know-about-nfls-new-bylaw-for-roster-management-in-2023>. The 2022 NFC Championship Game was played Jan. 29, 2023; Philadelphia won 31–7.

Watch for it on Sunday

💡 Watch for it: the cap check

Pick one team before its game (Over the Cap and Spotrac have free team pages) and do a five-minute cap check.

Before kickoff

- **Find the five biggest cap hits.** Are they at premium positions (QB, edge, left tackle, receiver, corner)? A top-five hit at running back, linebacker or safety is a team paying for a player, not a position.
- **Find the dead money.** A large number means a recent mistake or a recent release; it tells you which era of the roster the team is paying for.
- **Is the quarterback on a rookie contract?** If so, look for the team spending the savings: expensive veterans elsewhere, a deadline trade. If not, look for bargains and late-round starters around him.
- **Check the inactives** (posted 90 minutes before kickoff) and any **practice-squad elevations** announced the day before. They tell you the game plan's priorities and who is hurt.

During the game

- When a veteran with a big cap hit plays poorly, ask the cap question: what would releasing him save, and how much dead money would it leave? A small dead-money figure next to a big cap hit is a player likely to be released or asked to take a pay cut in March.
- When a young player on a rookie deal plays well at a premium position, you are watching the most valuable thing in football: his team's cap plan for the next three years.

The prediction: before the season ends, name the one contract the team will **restructure** (a star it plans to keep, with a large base salary next year) and the one it will **cut** (an older player at a lower-value position whose cap hit far exceeds his dead money). Check in March.

Predict the play

In this chapter the “play” is a front-office move. The pictures are a cap table, a trade offer and a scheme.

💡 Predict the play 1: the March squeeze

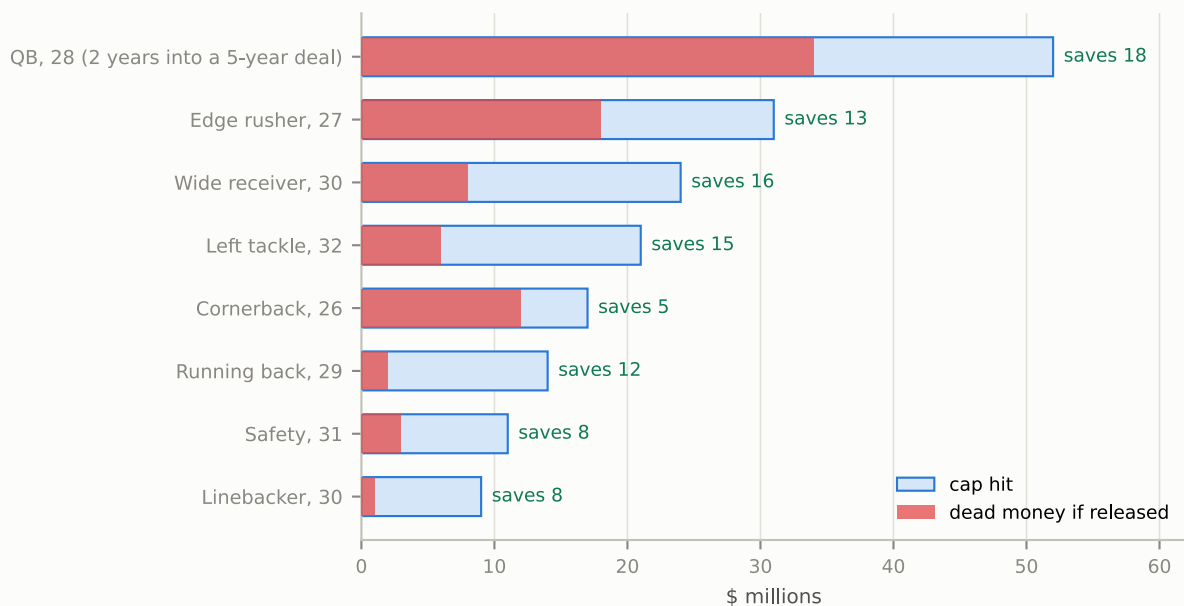


Figure 16: An invented team's eight largest cap hits for next season, in \$ millions (bar length), with the dead money each would leave if he were released before June 1 (red part of the bar) and the cap saved by releasing him (number at the right).

The team is \$14 million over next year's cap. Ages are next season's.

The team must clear \$14 million before free agency. Which one or two moves does it make, and which obvious move does it avoid?

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play 2: the trade-up

It is March. A team holding the 12th pick wants a quarterback and calls the team at No. 3. The offer, which is exactly what the 49ers sent Miami in 2021:

The 49ers gave	Johnson chart points	Surplus, % of cap per season
2021 No. 12	1,200	3.3%
2022 first-rounder (became No. 29)	640	2.5%
2022 third-rounder (became No. 102)	92	1.1%
2023 first-rounder (became No. 29)	640	2.5%
Total given	2,572	9.5%
They received: No. 3	2,200	3.4%

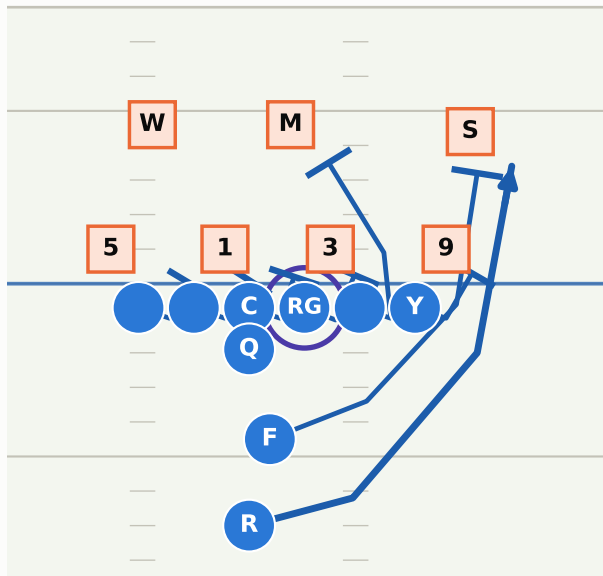
(Pick values use the picks' eventual slots; surplus values come from Figure 6.)

By each chart, who won the trade? And what would have to be true for the trade to be right anyway?

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: which guard fits?

Their bread-and-butter run



Prospect A

6-4, 334 lb, arms 34 in.
5.38 forty, 4.90 short shuttle
Film: drives defenders 3 yards
off the ball; slow to reach
and to climb to linebackers
RAS 6.1

Prospect B

6-4, 305 lb, arms 32.5 in.
5.02 forty, 4.45 short shuttle
Film: quick to reach and climb;
gives ground to big bull rushers
RAS 9.0

Figure 17: A team's most-called run (illustrative): outside zone from under center with a fullback, the line stepping right in unison. The team needs a right guard (ringed). Two prospects are on its board at the same pick.

Both prospects grade as starters. Which one does this team draft, and why might the other be the better pick for a different team?

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **Positional value is real, but price is not value.** The market pays quarterbacks most by far, then edge rushers, left tackles, receivers and corners. Yet a team's share of spending on any position barely predicts winning (r near zero, 2014–2025), while quarterback *play* predicts it strongly (r about 0.7). Value comes from getting play above your price.
- **A contract is a cap schedule, not a number.** Signing bonuses are paid up front but prorated over up to five years; restructures and void years borrow cap space from the future; releasing a player accelerates his remaining proration as dead money, which is why cutting a star can save almost nothing.
- **Every acquisition route has a currency.** Free agency costs cap space at peak prices; tags buy one year at a top-of-market price; trades cost picks and move contracts; compensatory picks reward letting players go; waivers and the practice squad are how teams find the 53rd man.
- **The draft is the cheapest source of good players, and the Johnson chart overprices the top.** Surplus value (production minus rookie cost) falls far more gently than the chart, so trading down usually buys value and trading up usually pays a premium, except, sometimes, for a franchise quarterback.
- **The rookie-contract window is the biggest edge in roster building.** A star quarterback on a rookie deal, like Purdy in 2022–2024, frees a fifth or more of the cap to spend on everyone else.
- **Scheme decides who is worth drafting.** Wide-zone and gap teams want different linemen, two-gap and one-gap fronts want different tackles, press and off teams want different corners. RAS and the combine filter for athleticism; film and fit decide the pick.

Connections

Builds on: Expected Points, EPA and Success Rate (EPA per dropback, and why passing outweighs running, the root of positional value); The Game Plan (matchup hunting, which only works with the players a front office bought); The 1978 Liberation (where free agency and the cap came from); Zone Running, Gap and Power Runs, DL Techniques and Gap Control and Coverage Fundamentals (the jobs behind scheme fit).

Sets up: Scheme and Fantasy: Reading Usage, Not Box Scores, which takes the same idea (what a scheme asks of each position) and turns it into fantasy usage: snap share, target share and the roles that predict points. The case studies show these choices in full: the Eagles paying for the lines, the 49ers and the Shanahan tree drafting for wide zone, and the Chiefs living through the end of a rookie-contract window. The Capstone puts the cap check into a full game-day routine.

Go deeper

For contracts, **Over the Cap** (overthecap.com, which also hosts the full collective bargaining agreement by article) and **Spotrac** are the standard free references; Jason Fitzgerald and Brad Spielberger's work at Over the Cap explains cap mechanics with real deals. Cade Massey and Richard Thaler's "The Loser's Curse" (*Management Science*, 2013) is readable and still the foundational paper on draft value. Andrew Brandt, a former Packers executive, writes about the business of football for *Sports Illustrated* and teaches it at Villanova. For scouting, Kent Lee Platte's ras.football explains RAS and lets you look up any prospect, and Pat Kirwan's *Take Your Eye Off the Ball* has a front-office view of how rosters are assembled. To extend this chapter's data, change the position groups in the first code cell, or swap "PHI" for any team in Figure 14.

Answers to Predict the play

Answer 1.

Release the running back (\$12 million saved) and restructure the quarterback; don't cut the left tackle.

Start with the cuts that are cheap *and* sensible. The running back saves \$12 million against only \$2 million of dead money, at the position the market values least, at an age when backs decline fast. That alone nearly solves the problem. The safety (\$8 million saved) and linebacker (\$8 million) are the same kind of move if the team needs more room, and both play lower-value positions.

Then the restructure. The quarterback is 28 and two years into a long deal: the team plans to keep him for years. Turning, say, \$25 million of next year's base salary into a signing bonus spread over his remaining three seasons lowers next year's hit by about \$16 million (the new bonus counts about \$8 million a year). It costs nothing on the field; it just borrows from later years, which is fine for a player you want on the team in those years.

The tempting mistake is the left tackle: 32 years old, \$15 million saved. But left tackle is a premium position, replacements are expensive, and the quarterback the team just restructured needs protecting. If the tackle is still good, cutting him to save money creates a hole that will cost more to fill. The edge rusher and cornerback are young, at premium positions and carry heavy dead money; they are extension candidates, not cuts. The wide receiver is the judgment call: \$16 million saved at 30 is real money, but a premium position, so a team would more likely ask him to take a pay cut than release him outright.

Answer 2.

By the Johnson chart, San Francisco overpaid by about 17 percent, a normal premium for a quarterback trade-up. By surplus value, it gave up about 2.8 times what it received.

The two charts tell different stories because they disagree about the slope. The Johnson chart thinks No. 3 is worth almost twice No. 12, so three picks for one looks only a little expensive. Surplus value thinks No. 12 and No. 29 are each worth a large fraction of No. 3, so giving three of them for one is a steep price.

What would make it right anyway? The quarterback exception. Average surplus is an average over every position; a franchise quarterback on a rookie contract is worth far more than the average pick, as Figure 3 showed. If the team is right that the player at No. 3 is a franchise quarterback, and the other picks wouldn't have produced one, the trade can pay for itself many times over. The surplus curve doesn't forbid trading up; it puts a price on the certainty you're claiming. In this case the certainty was wrong: Trey Lance was traded away for a fourth-round pick in 2023, and the 49ers' quarterback was the 262nd pick of the 2022 draft.

Answer 3.

Prospect B. Look at the diagram: every lineman's first step is sideways, and the right guard's job on this play is to reach a 3-technique who lines up outside him, then sometimes climb to a linebacker. That is a race, and B's quickness (a 4.45 short shuttle is excellent for a guard) and film ("quick to reach and climb") are exactly the traits it rewards. His weakness, giving ground to big bull rushers, is a real cost in pass protection, and this team will accept it because the scheme rarely asks him to stand still and absorb a 330-pound man head-on.

Prospect A is the better *gap* lineman: heavy, long, and able to move a defender off his spot on a down block, the power-team skill. On a gap team's board he may be the higher-rated player, and on this team's board he is a worse fit who would spend his first year trying to win races he isn't built for. That's scheme fit: the same pick, two right answers, depending on what the team runs. It also explains why a wide-zone team can often draft B a round later than his talent suggests. Fewer teams are shopping for him.